

DENY. DISRUPT. DISMANTLE.

Breaking the Business Model of Cybercrime in the
Gray Zone

Carole House | CEO, Penumbra **Strategies LLC**

WHERE I'VE SAT SHAPED WHERE I STAND



PENUMBRA

- Founder & CEO, Penumbra Strategies | CISSP
- Senior Fellow, Atlantic Council | Distinguished Senior Fellow, ACAMS International Anti-Fraud & Technology Task Force
- 3× White House (2× NSC, OMB)
- Executive Orders on emerging tech, cyber, and digital assets
- Regulatory advisor (CFTC, New York, Idaho) + California Governor's Innovation Council



1LT Carole House
Regulator 6
2nd PLT, 45th MICO, 2-1 CAV
March 2013 - April 2014



THE CORE ARGUMENT

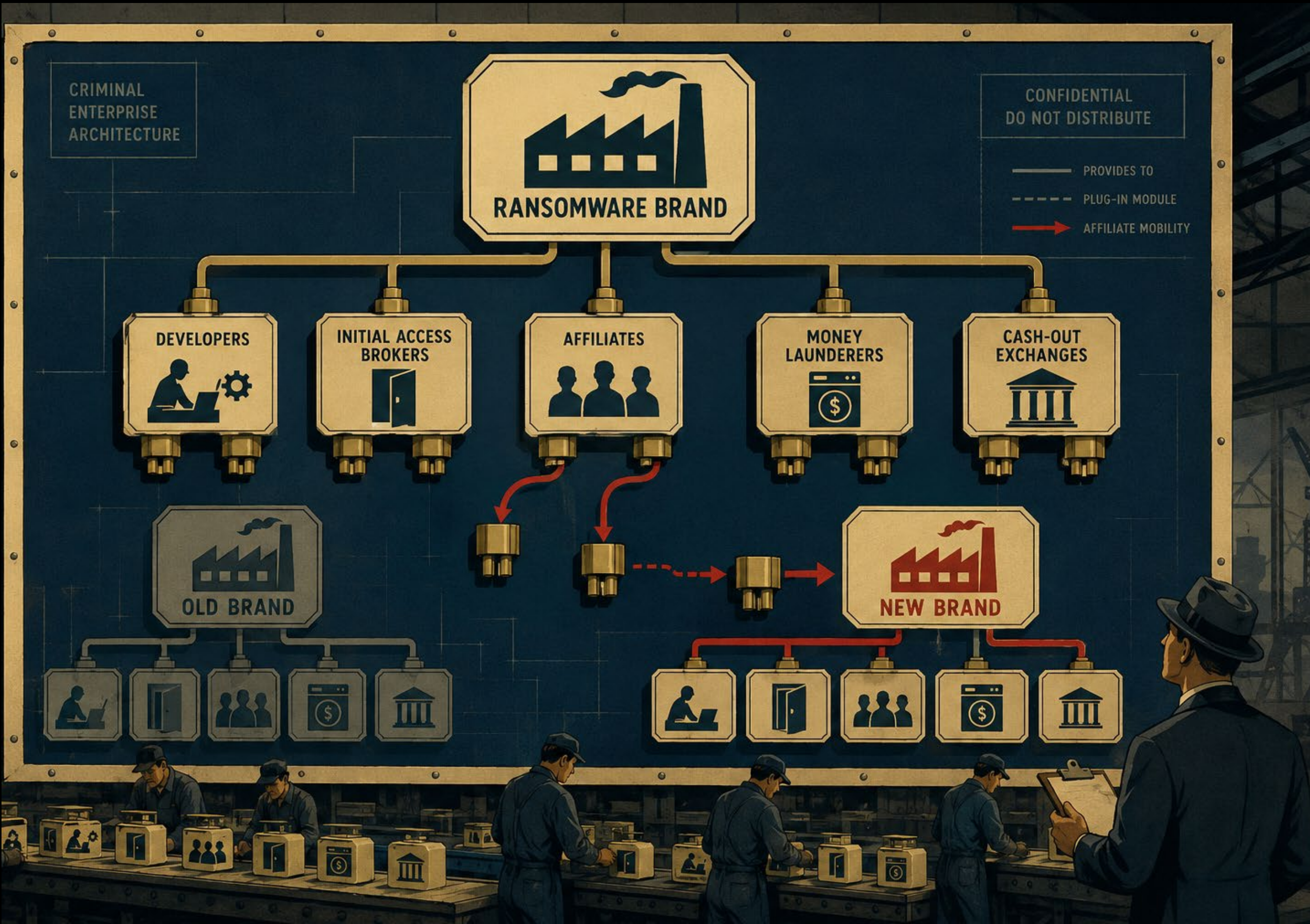
Cybercrime networks – including ransomware transnational criminal organizations (TCOs), pig-butcher scam compounds – are not mere criminals to be prosecuted. They are transnational organized crime ecosystems with state-level consequence-generation capacity, requiring campaign planning logic and lessons from asymmetric warfare to be applied.

Military campaign planning doctrine provides specific analytic tools for the problems counter-cybercrime police most consistently fails to solve.

The U.S. accelerated a genuine paradigm shift in 2021, applying whole-of-government campaign thinking to ransomware, which produced some real and some ephemeral results. A doctrinal lens can help us evaluate what the next evolution requires.

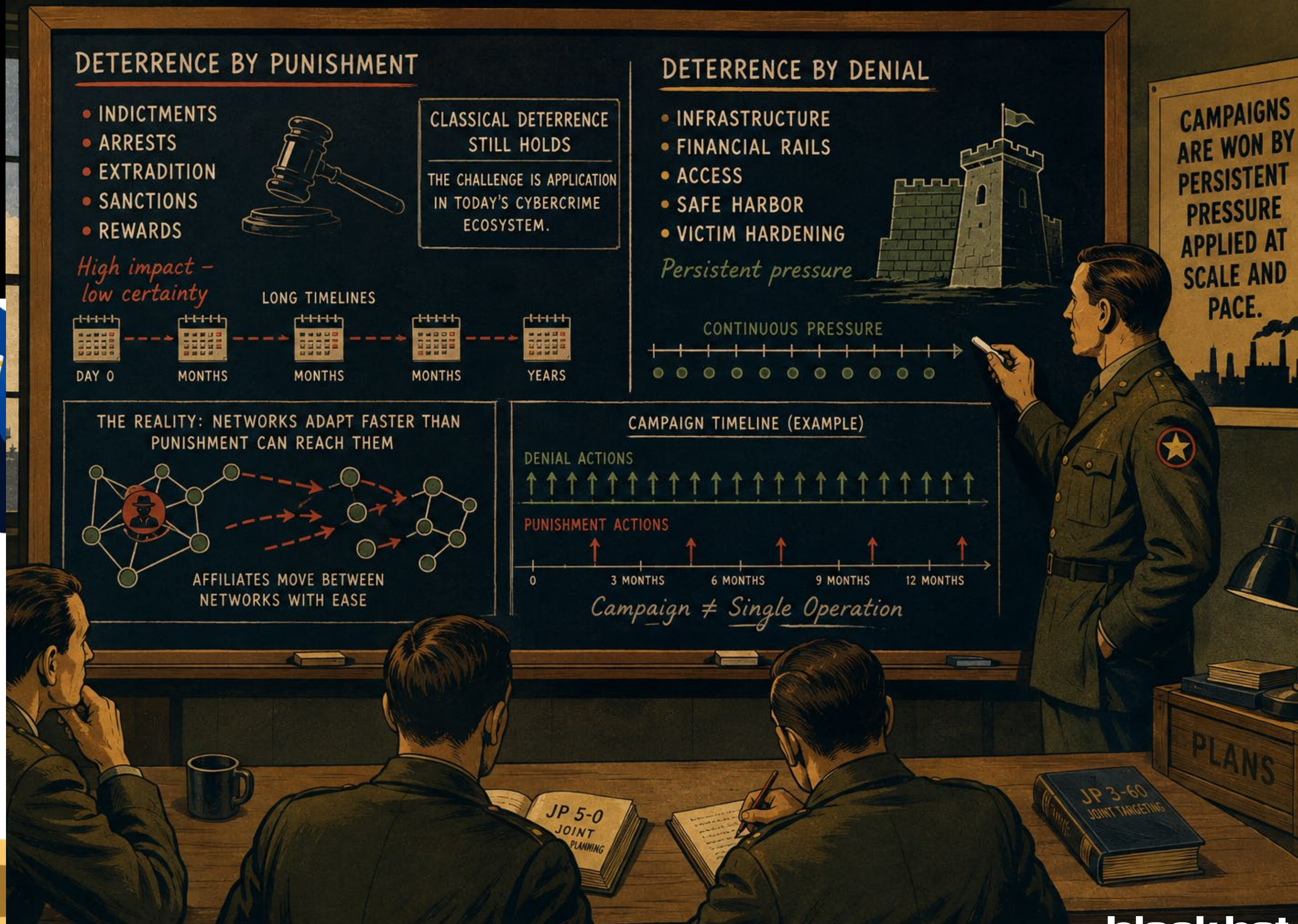
Disruption operations are not offensive adjuncts to a defensive strategy. They are the precondition for defense to compound.

UNMATCHED INDUSTRIALIZATION CREATES A DETERRENCE PROBLEM

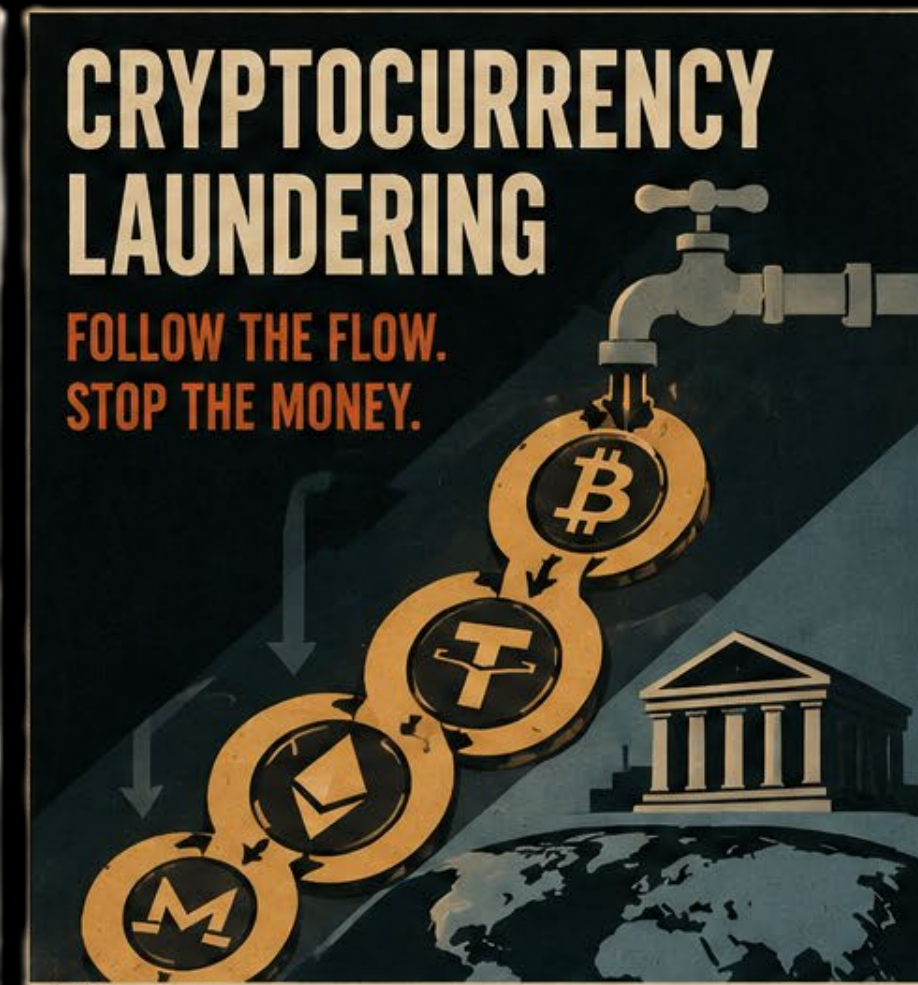
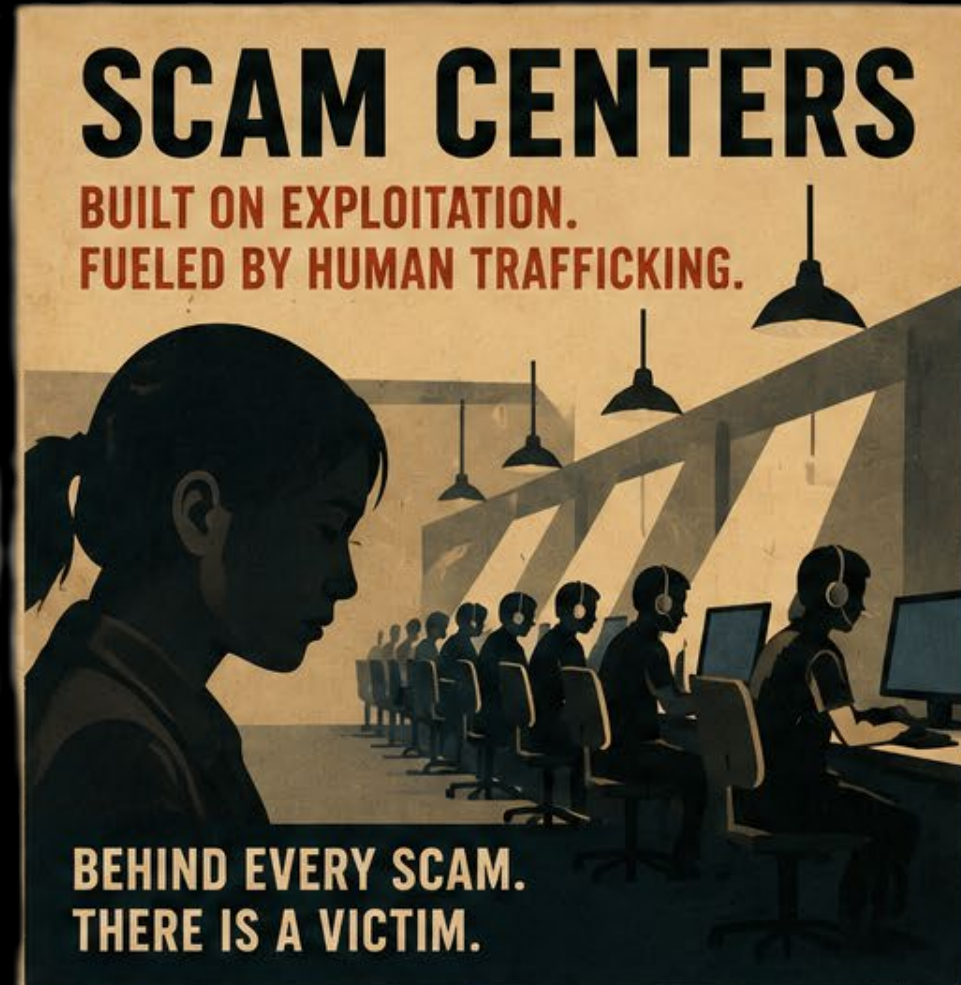


- Traditional criminal activity lens → the adversary outpaces the model
- Asymmetric warfare lessons → non-state actor with state-level effects
- Distinctions can determine: which authorities apply, which agencies lead, how you target, how measure success

LEGACY DETERRENCE HAS BEEN MISMATCHED AGAINST CYBERCRIME ECOSYSTEMS



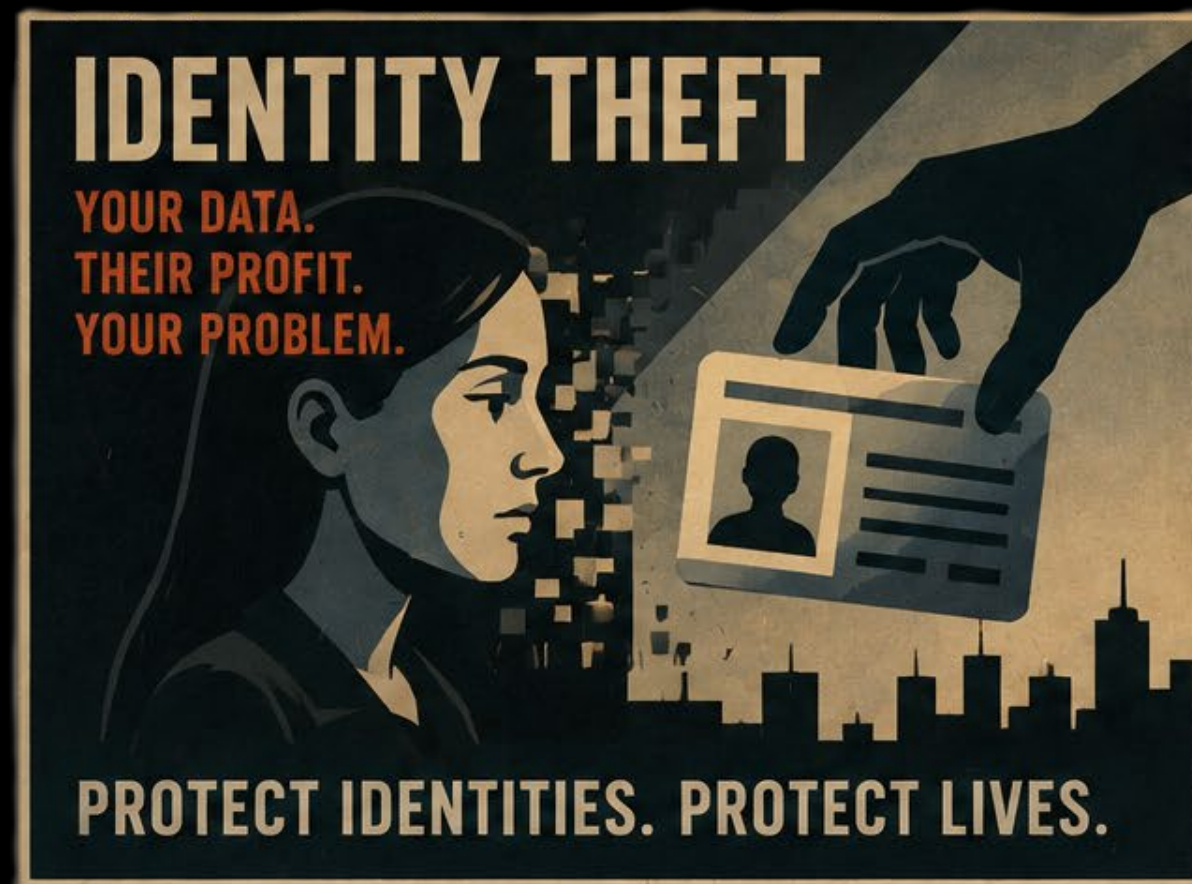
MOST (CYBER)CRIME IS FINANCIAL CRIME



Beyond buzzwords --- the journey to prioritization:

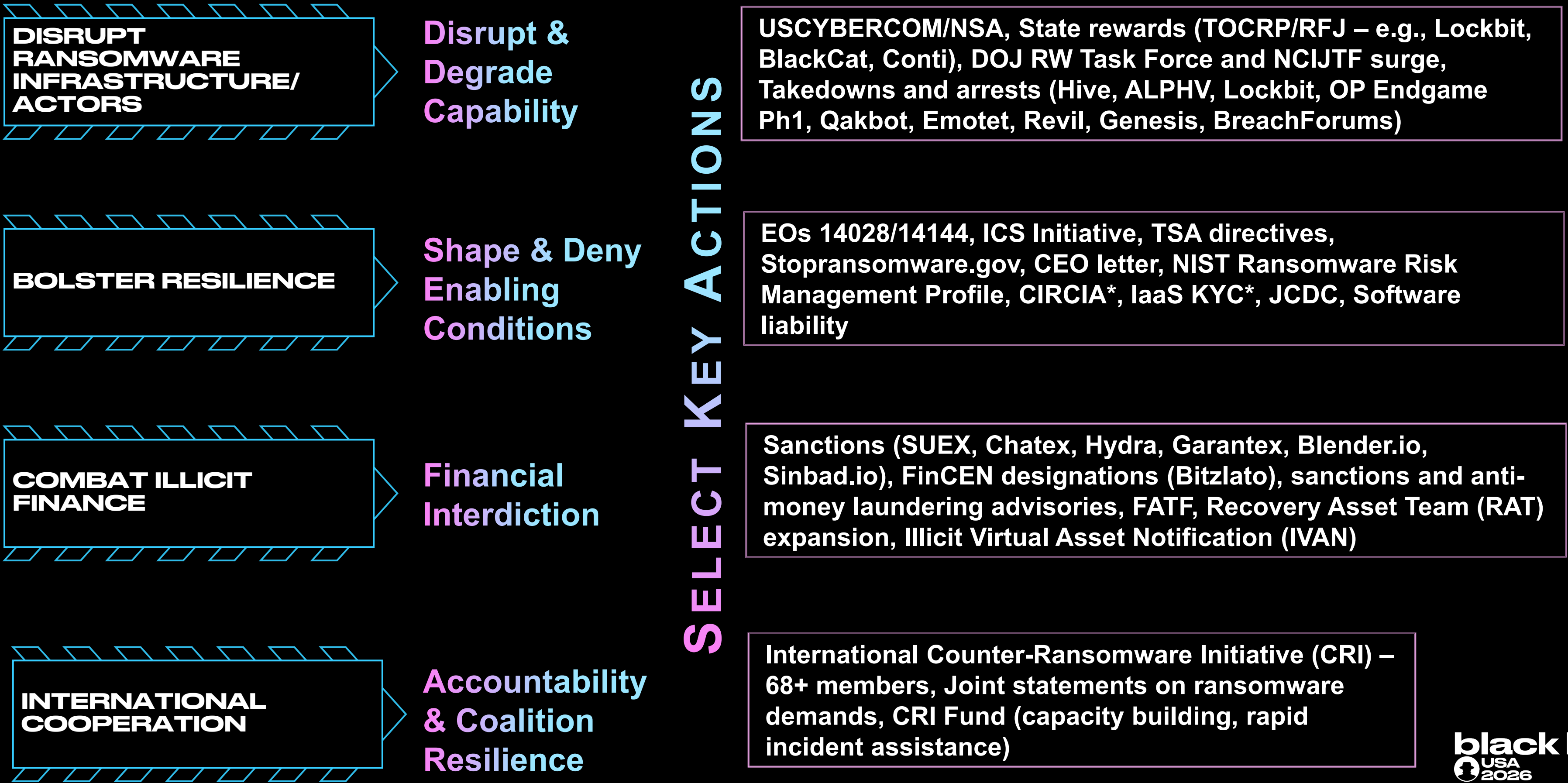
- Ransomware's disruptive nature
- Scope and scale of frauds and scams

What solves one of these tends to help the others:



- Identity compromise at the heart of most cyber- and financial- crime
- Cryptocurrency unique (but not inevitable) role across cybercrime-as-a-service

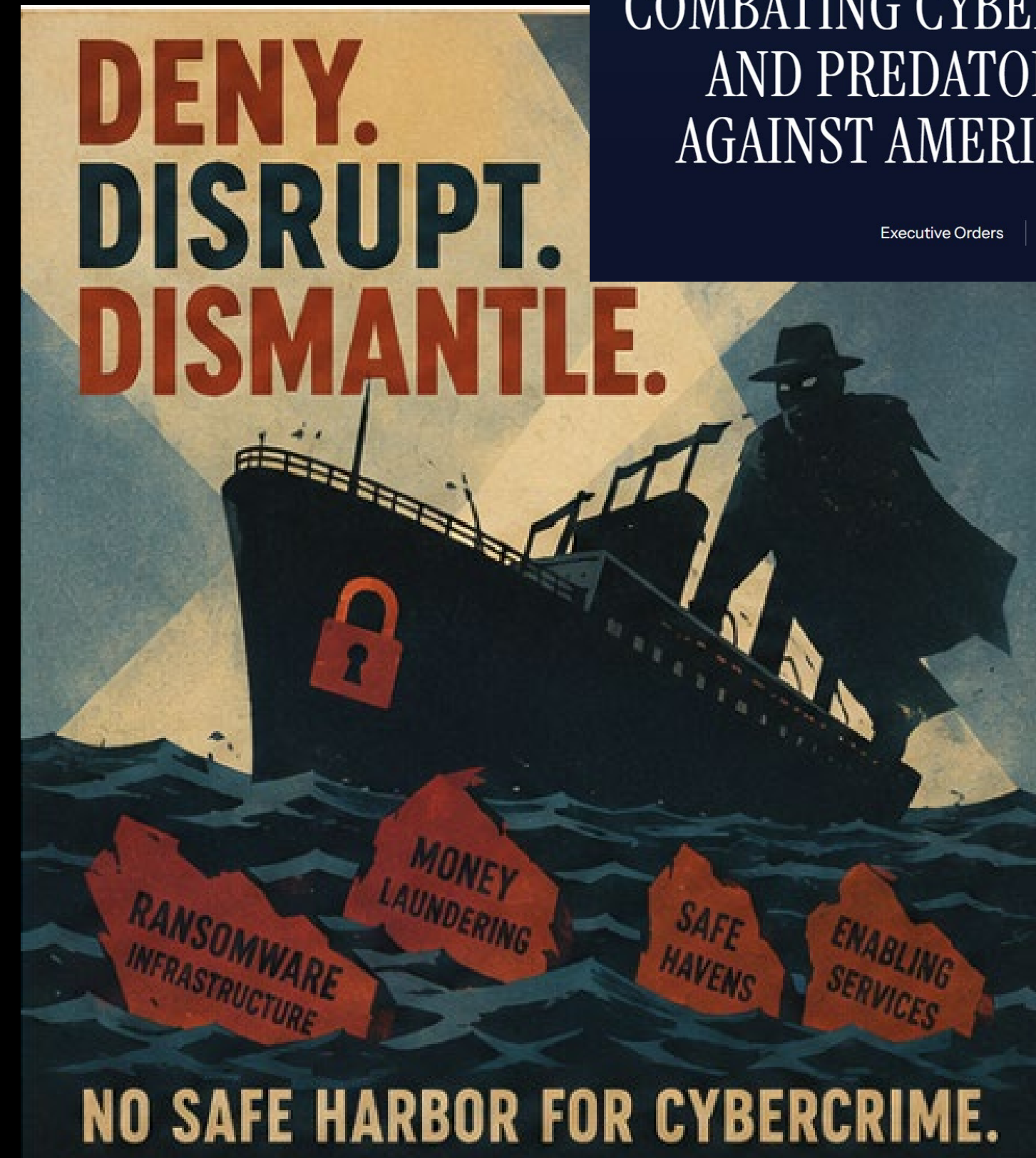
THE PARADIGM SHIFT: THE U.S. COUNTER-RANSOMWARE CAMPAIGN (2021-2025)



AN OPPORTUNITY NOW TO BUILD THE NEXT CAMPAIGN

Executive Order 14390 on Combating Cybercrime, Fraud, and Predatory Schemes against American Citizens (March 2026)

- 60-day review of USG tools
- 120-day Action Plan – identify TCOs, propose solutions, create operational cell within National Coordination Center (NCC), industry partnership
- AG to prioritize prosecutions and recommendations for Victims Restoration fund
- Sec. State to impose sanctions, visa restrictions, foreign aid cuts



PRESIDENTIAL ACTIONS

COMBATING CYBERCRIME, FRAUD,
AND PREDATORY SCHEMES
AGAINST AMERICAN CITIZENS

Executive Orders

March 6, 2026

MILITARY PLANNING AND TARGETING FRAMEWORKS: A DIAGNOSTIC LENS



Analytical and planning tools for sequencing, targeting, and assessing operations against network-structured adversaries.

Non-cyber solutions to cyber problems...

START WITH THE END(STATE)

Strategy = Ends + Ways + Means

Defining what we want – observable strategic effect
(e.g., *ransom revenue falls? Safe harbor cooperation improves? Recruitment declines? Critical infrastructure incident [disruptive? financial?] impacts reduced?*)

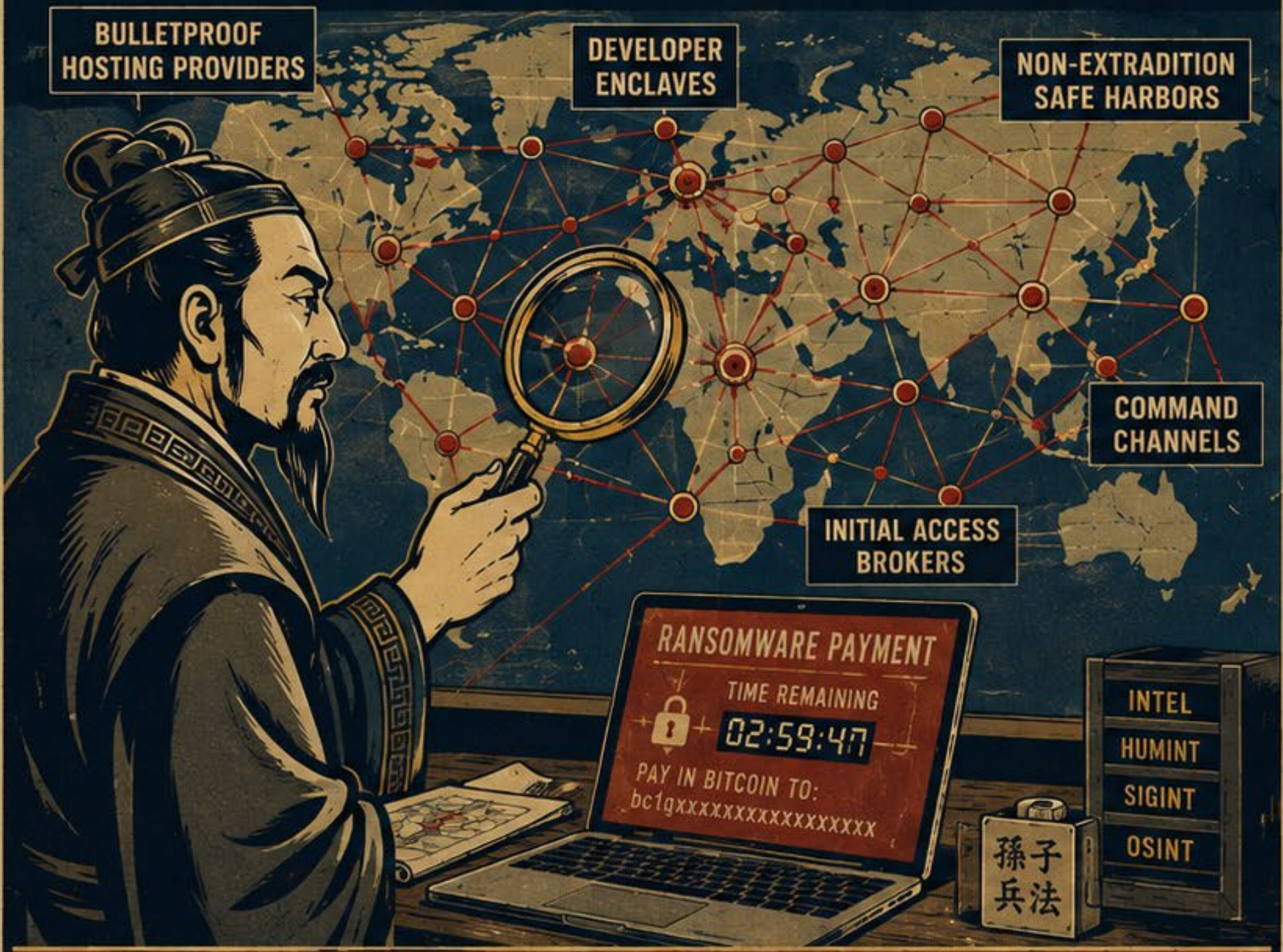
- Sustained, but time-bounded
- Measurable



Ref. COL (Ret.) Arthur F. Lukke, US Army, Military Review

LEARNING FROM *THE ART OF WAR*/ON WAR

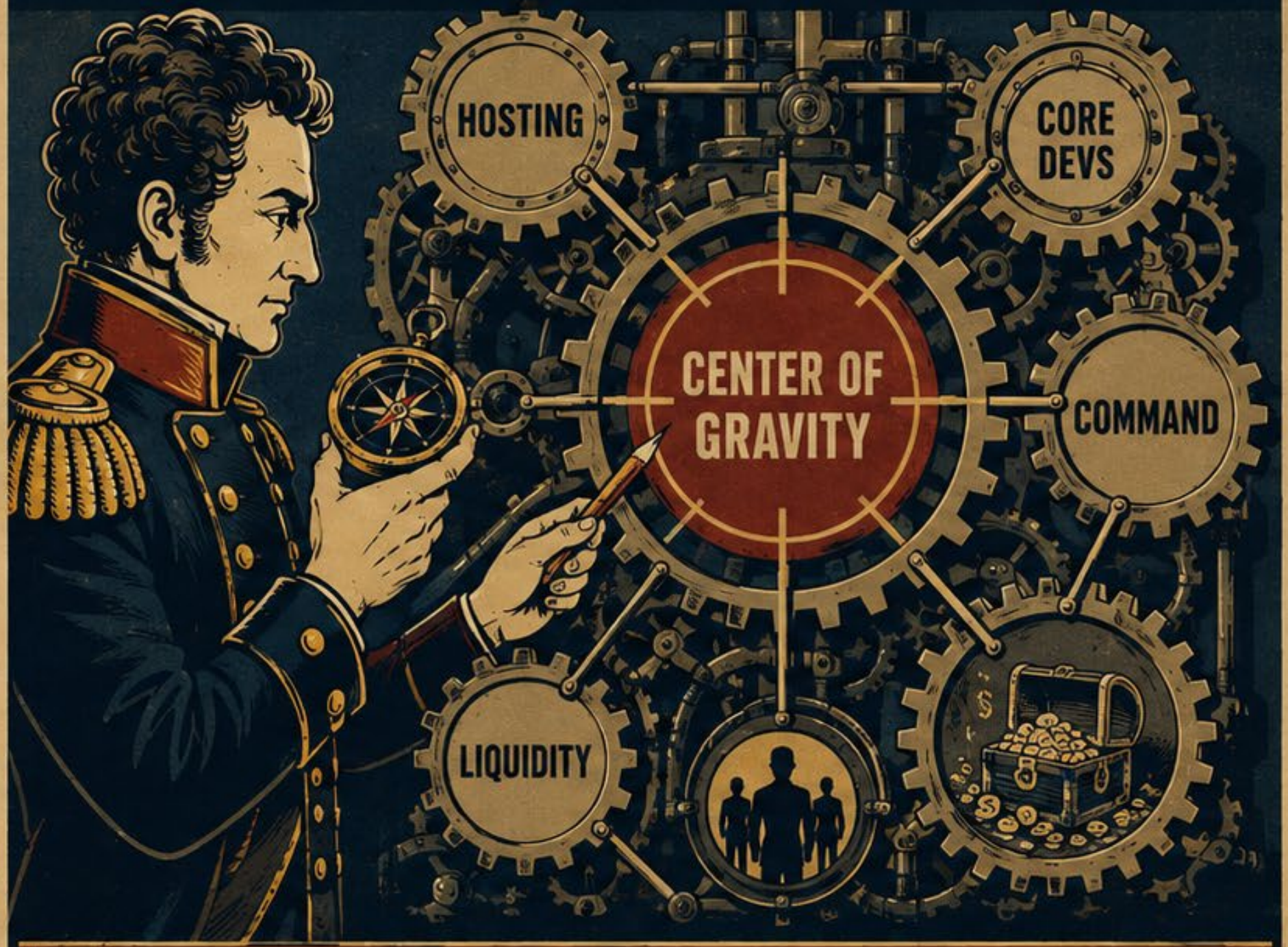
1. MAP THE ECOSYSTEM



KNOW YOUR ENEMY AND KNOW YOURSELF.

Target System Analysis/Development and Prioritization – Organize around priority networks (ex. Standardized Approach for Banding Ransomware [SABR])

2. INTERDICT THE COG



STRIKE THE HUB OF ALL POWER AND MOVEMENT.

CoG System Analysis – e.g., key financial, technical, and state patronage nodes

TARGETING NETWORKS

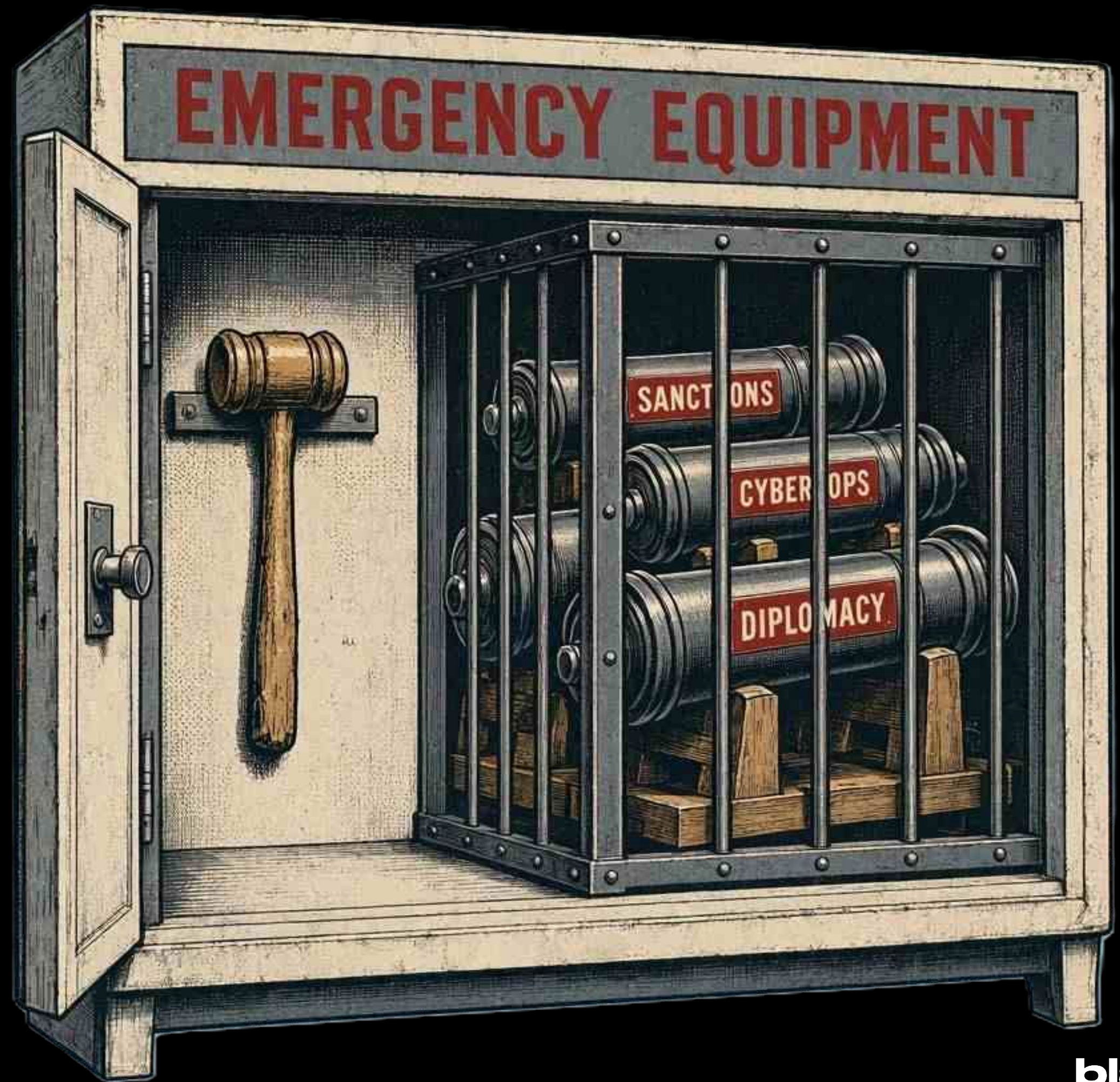
DECAPITATION WORKS WHEN	DECAPITATION FAILS WHEN	DEGRADING NETWORKS
Hierarchical structure	Modular and fungible	Simultaneous multi-node interdiction
Simultaneous infrastructure pressure	Intact enabling conditions	Trust network poisoning
Degraded viability threshold	Fast migration	Financial and reconstitution denial



Ref. Sparrow (1991) · Morselli (2009) · Bouchard (2020) · Bright et al. (2017, 2020)

CHOOSING YOUR TOOL...

(AND WHY IS IT ALWAYS SANCTIONS?)



THE TOOLKIT: INSTRUMENT FEATURES <i>(Illustrative)</i>					
Tool/Instrument	Primary Effect	Resource Intensity (Time/Friction to USG)	Resistance to Reconstitution	Durability/Cost Imposed on Adversary	Key Dependencies and Vulnerabilities
OFAC Sanctions & Financial Designations	Ecosystem-wide chilling effect & financial isolation	LOW–MEDIUM	MEDIUM	HIGH if compliant rails; LOW in non-cooperative safe harbors	- Compliant jurisdiction enforcement - Vulnerable without parallel int’l or LE/Cyber action (e.g., Garantex gap)
Financial Interdiction Partnerships (IVAN, RAT)	Real-time payment/asset interception in transit	LOW-MEDIUM	MEDIUM	MEDIUM-HIGH <i>(Immediate cash-flow hit, but could be rerouted)</i>	- Mandatory reporting triggers (CIRCI^A*, SARs) - Compliant VASP/banking participation
Cyber Operations & Infrastructure Takedowns	Immediate capability denial & operational disruption	MEDIUM-HIGH	LOW	LOW–MEDIUM alone; VERY HIGH if paired with supporting financial or network disruption	- F3EAD - design the info gain before the takedown - Requires concurrent financial or personnel disruption to prevent instant rebuilding
DOJ Prosecutions & Indictments	Permanent actor removal & chilling effect on recruitment	VERY HIGH	HIGH	HIGH for convicted individuals; LOW for fungible low-level affiliates	- Foreign jurisdiction cooperation & extradition; USG enforcement apparatus working; high attribution - Most effective as consolidating action on an already degraded network
Diplomatic Pressure & CRI Coalitions	Safe harbor elimination & partner LE enablement	LOW–HIGH	MEDIUM-HIGH	VARIABLE <i>(High structural friction when aligned, fragile under geopolitical shifts)</i>	- Sustained trade, financial, or political leverage - Vulnerable to broader geopolitical ruptures (e.g., REvil post-Ukraine)
Regulatory Action (CIRCI ^A , AML/CFT, FinCEN 311)	Ecosystem hardening, intelligence generation, & structural denial	VERY HIGH	HIGH	PERMANENT/STRUCTURAL <i>(Can permanently alter adversary op’l economics)</i>	- Legislative/regulatory follow-through - Private sector compliance and reporting adoption

THE PERENNIAL SIDE CHAINS: TWO POLICY DEBATES THAT MISS THE TARGET (BUT UNDERSTANDABLY ALWAYS GET RAISED)

Correct instinct

- ✓ Reduce ransomware profitability

Strategic risk

- Enforcement shifts toward victims
- Discourages reporting
- Weakens intelligence
- Doesn't dismantle laundering infrastructure



FOCUSING HERE BURNS RESOURCES,
DISCOURAGES REPORTING, AND
LEAVES THE MACHINE RUNNING.



FRAGMENTS AUTHORITY, INCREASES RISK,
REDUCES ACCOUNTABILITY,
AND UNDERMINES CAMPAIGN UNITY.

Correct instinct

- ✓ Government response has not scaled

Strategic risk

- Fragments use of force
- Escalation and attribution risks
- Weak accountability
- Replaces (not reinforces) or conflicts with whole-of-government campaigns

ANOTHER QUICK SIDE-CHAIN: THE INFORMATION SHARING DEBATE – OPPORTUNITIES TO EXPAND

Information Sharing to Combat Fraud

Analysis of Statutory Permissions and Constraints

ACAMS

GASA

July 2026

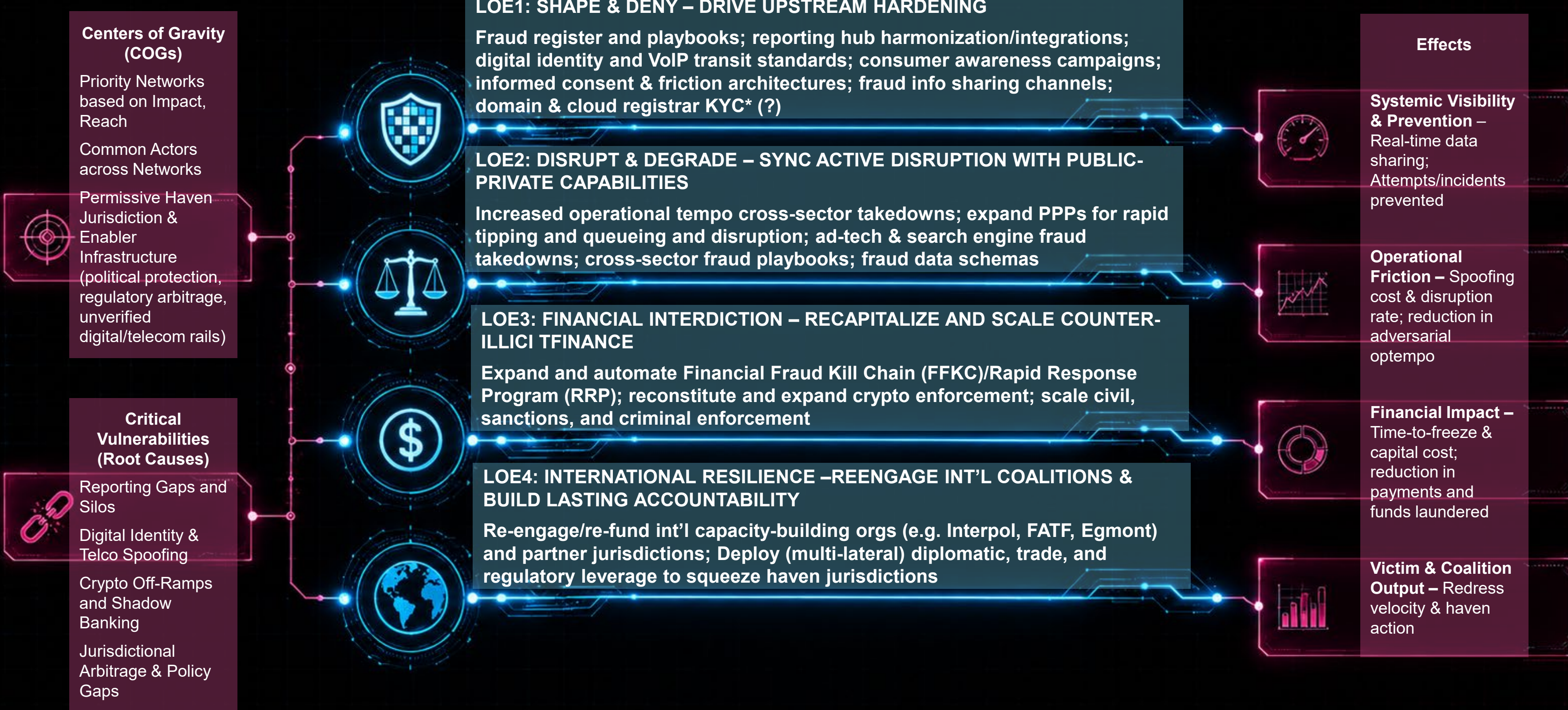
Prepared by ACAMS and the Global Anti-Scam Alliance (GASA)

Use Case	CISA 2015	§ 314(b)	ECPA (Wiretap/SCA)	CCPA	GLBA	FCRA	EFTA / RFPA / BSA	GDPR
Mule Account ID & Sharing <i>Bank → Bank; account identifiers & transaction patterns; real-time interdiction</i>	Genuinely Unsettled	Permitted (Safe Harbor)	Conditionally Permitted	Conditionally Permitted	Permitted	Not Applicable	Conditionally Permitted	Conditionally Permitted
Scam Phone Number ID & Blocking <i>Telco → Consumers / Tech Platforms; phone numbers, complaints, call patterns; pre-event / near-real-time prevention</i>	Conditionally Permitted	Not Applicable	Conditionally Permitted	Conditionally Permitted	Not Applicable	Not Applicable	Not Applicable	Conditionally Permitted
Prosecution & Accountability Support <i>Any sector → Law Enforcement; evidentiary data & case files; post-event prosecution</i>	Conditionally Permitted	Not Applicable	Restricted	Permitted	Permitted	Conditionally Permitted	Permitted (Safe Harbor)	Conditionally Permitted

Free/publicly available ACAMS International Anti-Fraud & Technology Task Force + Global Anti-Scams Alliance (GASA) + Akin Gump LLP analysis to help counter-fraud efforts

Assess Effects with Honesty and Candor					
Action or Operation	Duration	Scope	Operational Effects	Financial Effects	Reconstitution Friction
Hive (2022–23)	✓ 6 months pre-action exploitation ✗ Affiliates migrated within months	✓ 1,300+ victims served	✓ Intelligence-driven disruption; network degraded before takedown ✗ Affiliate network reconstituted elsewhere	✓ \$210M+ victim losses averted	✓ Affiliate confidence degraded
Garantex (2022–2025)	✓ Sanctioned Apr 2022; dismantled Mar 2025 when server infrastructure reached in cooperative jurisdictions ✗ Processed ~\$96B despite designation	✓ Raised sanctions compliance expectations across VASPs ✗ Successor exchange Grinex rapidly emerged	✓ Demonstrated benefits of int’l partner cooperation to dismantle tech infrastructure ✗ Continued illicit activity post-designation	✓ Financial isolation when jurisdictions cooperate ✗ Designation alone insufficient without cooperative jurisdiction and crypto sector	✓ Enduring policy precedent ✗ Successor infrastructure required additional enforcement
Op Cronos/ LockBit (Feb 2024)	✓ 79% payment decline ~6 months (Chainalysis 2024) ✗ Reportedly back in 5 days and substantially rebuilt ~18 months	✓ 10 countries; Panev extradited Mar 2025	✓ Khoroshev identified; affiliate confidence degraded ✗ Affiliates reportedly moved to RansomHub	✓ Disruption of criminal financial operations; \$10M RFJ reward ✗ Financial pressure not synchronized	✓ Khoroshev exposure = persistent travel risk ✗ Financial infrastructure largely remained intact
Op Endgame (2024–25)	✓ 3 iterative seasons; S1 IOCs fed S2 ✗ Successor tooling (reportedly) emerging	✓ Initial access broker (IAB) infrastructure across multiple RaaS groups	✓ Developer and malware-loader indictments; multiple rounds ✗ Limited concurrent financial interdiction	✓ EUR 21.2M seized	✓ Adversary forced into constant retooling
Prince Group (2024–26)	✓ 146 designations; US-UK coordinated; Section 311 on Huione ✗ Compounds relocating not shutting	✓ OFAC + FinCEN + DOJ + UK OFSI simultaneous ✗ State patronage COG largely unaddressed	✓ Coordinated sanctions on key facilitators; Huione severed from U.S. system ✗ Underground banking unaddressed	✓ Major laundering infrastructure isolated ✗ Successor payment infrastructure rapidly emerged	✓ Chinese enforcement demonstrated potential leverage ✗ Chen Zhi in Chinese (not U.S.) custody*
**This table is not comprehensive, is for discussion purposes, and aims to provide a lens for thinking about how to evaluate campaigns targeting cyber TCOs Black Hat USA 2026 17					

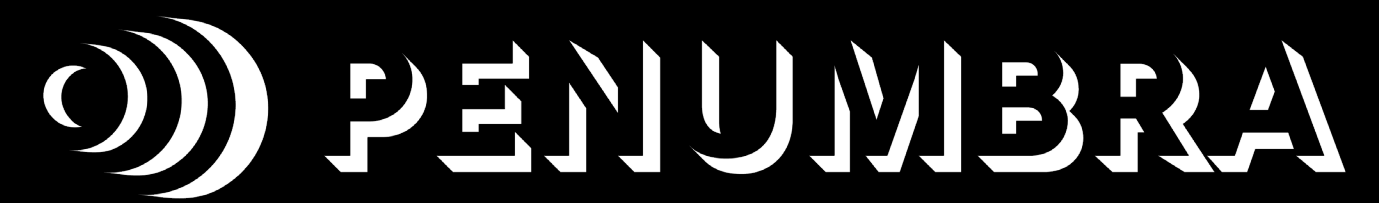
APPLYING THE FRAMEWORK TO A CYBER TCO



WHAT THE EO 14390 ACTION PLAN SHOULD DELIVER: FOUR STRATEGIC IMPERATIVES

1	Establish Co-Equal National Security Prioritization & Unified Targeting	<i>JP 5-0: End State + Priority Targeting</i>
2	Synchronize Capability Disruption with Aggressive Financial Interdiction	<i>JP 5-0 + F3EAD: Concurrent LOEs, Sustained Tempo</i>
3	Re-Engage, Expand, and Institutionalize International Coalitions	<i>LOE 4: Coalition Permanence & Partner Capacity</i>
4	Commit to the Build Phase – Deny Reconstitution & Restore Defenses	<i>FM 3-24 Build Phase: Deny Reconstitution</i>

Thank you



**DENY.
DISRUPT.
DISMANTLE.**



NO SAFE HARBOR FOR CYBERCRIME.

**BUILD RESILIENCE.
REDUCE VICTIMS.**



FOLLOW THE MONEY.



**DISRUPT THE FINANCIAL RAILS
THAT POWER CYBERCRIME.**

TOGETHER, WE ARE STRONGER.



**68 NATIONS. ONE MISSION.
A GLOBAL FRONT AGAINST RANSOMWARE.**

SPEED IS STRATEGY.



**OUTPACE. OUTADAPT. OUTLAST.
WIN IN THE OODA LOOP.**